Lesson 29: Differentiation and Parametric Equations

1) Parametric Equations: $x = t^2, \quad y = 4 - 7t$

a) Find $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} =$? _____ (multiple choice)

A) $\dfrac{-7}{t}$ B) $\dfrac{-7}{2t}$ C) $\dfrac{-7}{t^2}$ D) $\dfrac{-7}{5t}$

b) Find $\dfrac{d^2y}{dx^2} = \dfrac{\dfrac{d}{dt}\left[\dfrac{dy}{dx}\right]}{dx/dt} =$? _____ (multiple choice)

A) $\dfrac{-7}{(2t)^3}$ B) $\dfrac{7}{(2t)^3}$ C) $\dfrac{14}{(2t)^3}$ D) $\dfrac{10}{(2t)^3}$

2) Parametric Equations: $x = \sqrt{t}, \quad y = 5 - 3t$

a) Find $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} =$? _____ (multiple choice)

A) $-3t^{1/2}$ B) $-4t^{1/2}$ C) $-5t^{1/2}$ D) $-6t^{1/2}$

b) Find $\dfrac{d^2y}{dx^2} = \dfrac{\dfrac{d}{dt}\left[\dfrac{dy}{dx}\right]}{dx/dt} =$? _____ (multiple choice)

A) -3 B) -4 C) -5 D) -6

3) Parametric Equations: $x = 5t^2$, $y = 5 - 2t$

Find an equation of the tangent line at the point (5, 3).

a) Find $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} =$? _____ (multiple choice)

A) $\frac{-3}{5t}$ B) $\frac{-1}{5t}$ C) $\frac{-4}{5t}$ D) $\frac{-7}{5t}$

b) At the point $(5,3)$, $t =$? _____ Hint: $t = \sqrt{\frac{x}{5}}$

c) Equation of tangent line at $(5,3)$: ? _____

Hint: $y - y_1 = m(x - x_1)$ Equation of line formula

4) Parametric Equations: $x = t^2 + 2t - 4$, $y = 2t$

Find an equation of the tangent line at the point (-1, 2).

a) Find $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} =$? _____

A) $\frac{2}{2t^2+2}$ B) $\frac{2}{2t^2+2t}$ C) $\frac{2}{2t+2}$ D) $\frac{2}{2t-4}$

b) At the point $(-1,2)$, $t =$? _____ Hint: $t = \frac{y}{2}$

c) Equation of tangent line at $(-1,2)$: ? _____

Hint: $y - y_1 = m(x - x_1)$

5) Parametric Equations: $x = 3\cos t,\ \ y = 3\sin t \quad 0 \le t \le 2\pi$

Find an equation of the tangent line at the point (0, 3).

a) Find $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} =$? _____ (multiple choice)

A) $-\dfrac{\cos t}{\sin t}$ B) $\dfrac{\cos t}{\sin t}$ C) $-\dfrac{\sin t}{\cos t}$ D) $\dfrac{\sin t}{\cos t}$

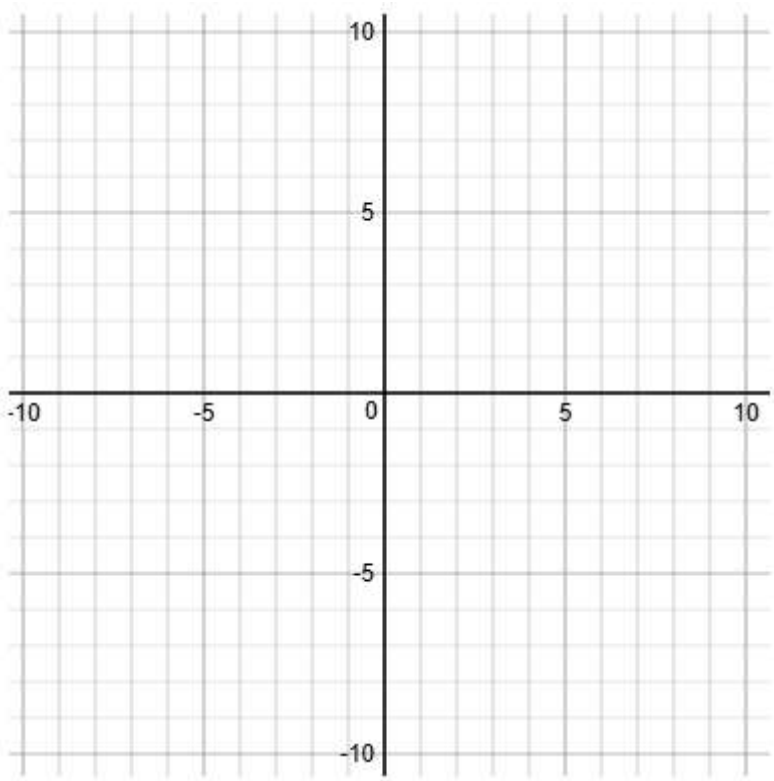
b) At the point $(0,3)$, $t =$? _____

Hint: $\cos t = \dfrac{x}{3} \Rightarrow t = \cos^{-1}\left(\dfrac{x}{3}\right)$

c) Equation of tangent line at $(0,3)$: ? _____

Hint: $y - y_1 = m(x - x_1)$

d) Draw the graph of $x(t) = 3\cos t$, $y(t) = 3\sin t$ and the graph of the tangent line at $(0, 3)$.



6) Parametric Equations: $x = 3t + t^2$, $y = 4t^{3/2}$

Find an equation of the tangent line at the point(0, 0).

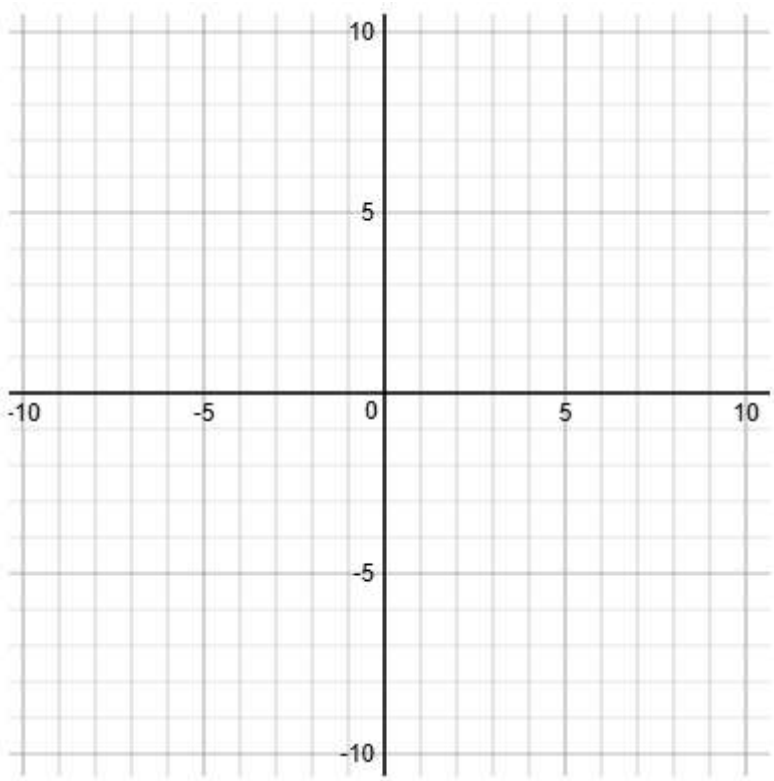
a) Find $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} =$? _____

A) $\dfrac{8t^{1/2}}{3+2t}$ B) $\dfrac{7t^{1/2}}{3+2t}$ C) $\dfrac{6t^{1/2}}{3+2t}$ D) $\dfrac{5t^{1/2}}{3+2t}$

b) At the point $(0,0)$, $t =$? _____

Hint: $t^{3/2} = \dfrac{y}{4} \Rightarrow t = \left(\dfrac{y}{4}\right)^{2/3}$

c) Equation of tangent line at $(0,0)$: ? _____

Hint: $y - y_1 = m(x - x_1)$

d) Draw the graph of $x(t) = 3t + t^2$, $y(t) = 4t^3$ and the graph of the tangent line at $(0, 0)$.